

the claim: "the minimum is m "

UPPER BOUND

"a program of size m exists"

WITNESSED

circuit (the object)

Layer 1 re-execution
 μs · shares no code with search

Layer 2 symbolic equivalence
 $\forall x. \text{circuit}(x) = Mx \cdot Z3$

Layer 3 proof term
Lean 4 kernel · axioms: [propext]

machine-checked certificate

the
**REFUTATION
GAP**

LOWER BOUND

"no program of size $m-1$ exists"

NOT WITNESSED

(no object exists)

SAT solver returns UNSAT
 10^4+ lines of C: inprocessing,
clause deletion, variable elimination

nothing to re-execute,
nothing to check

an unaudited assertion